N11 Search Load Test Scenarios

These test scenarios include coverage of the written load test scenarios of N11 Header Search Bar Module. This load test's aim is to control the behaviour and response of the search function, and it verifies the n11 search function's stability.

Test Environment : https://www.n11.com/
Module : N11 Header Search
Test Tool : Locust

TEST SCENARIOS

TS 01 - Search With Category Name

Description : It verifies that when a common category name is searched response with expected time and shows related category products properly.

Test Input : "bilgisayar"

Steps :

1. Go to home page
2. Search with the test input which is "bilgisayar"
3. Check the response time for `GET /arama?q=bilgisayar`
4. Control the response time and the result.

Expected Result:

1. HTTP 200 code should be returned.
2. Response time should be less than 2000ms.

TS02 - Search With Exist Product Name

Description : It verifies that when the existing products in the web site is searched, the query should be worked correctly in expected response time, and shows the related product.

Test Input : " Lenovo IdeaPad Slim 3 15IAN8"

Steps :

1. Go to home page
2. Search with the test input which is **" Lenovo IdeaPad Slim 3 15IAN8"**
3. Check the response time for `GET /arama?q=Lenovo+IdeaPad+Slim+3+15IAN8`
4. Control the HTTP status code

Expected Result:

1. When “Lenovo IdeaPad Slim 3 15IAN8” is searched, the existing product should be seen in 2000ms.

TS03 - Search With Random Keywords

Description : It verifies system response time when the random keys used for the search.

Test Input : “lfkjh”

Steps :

1. Go to home page
2. Search with the test input **“lfkjh**”
3. Measure response time for `GET /arama?q=lfkjh`
4. Validate HTTP status code
5. Check the "No results found" page is returned

Expected Result:

1. The system should gracefully return a "No results found" (Sonuç bulunamadı) page.
2. 2. HTTP 200 OK status should still be returned (not an error page).

TS04 - Search With Special Characters

Description : It verifies system behaviour, when the special characters used for the searching, without any system crashes.

Test Input : “‘^+%&/()@é”

Steps :

1. Go to home page
2. Search with the test input **“‘^+%&/()@é”**
3. Check response time for `GET /arama?q=^+%&/()@é`
4. Control the HTTP Status.

Expected Result:

1. HTTP 200 is returned with no errors
2. Special characters are correctly URL encoded

TS05 - Search With Empty Input

Description : It verifies system behaviour, when search is performed with an empty input, causing any server errors or any problems.

Test Input : “ “

Steps :

1. Go to home page
2. Search with the test input with space **“ ”**
3. Check the result and the system behaviour.

Expected Result:

1-The system should either show validation feedback or redirect to a default search page without errors.
2. No Server errors should occur.

TS06 - Search With Long Character Texts

Description: It verifies system stability and response when a very long search is made.

Test Input: 255 + character text

Steps:

1. Go to home page.
2. Search with the long test input.
3. Check if the system handles the request without any server problems.

Expected Result:

1-The system returns with a "no results" page.

2- HTTP response should be without an error code

TS07 - Pagination Search

Description: It verifies that search result can be navigated.

Test Input: “araba”

Steps:

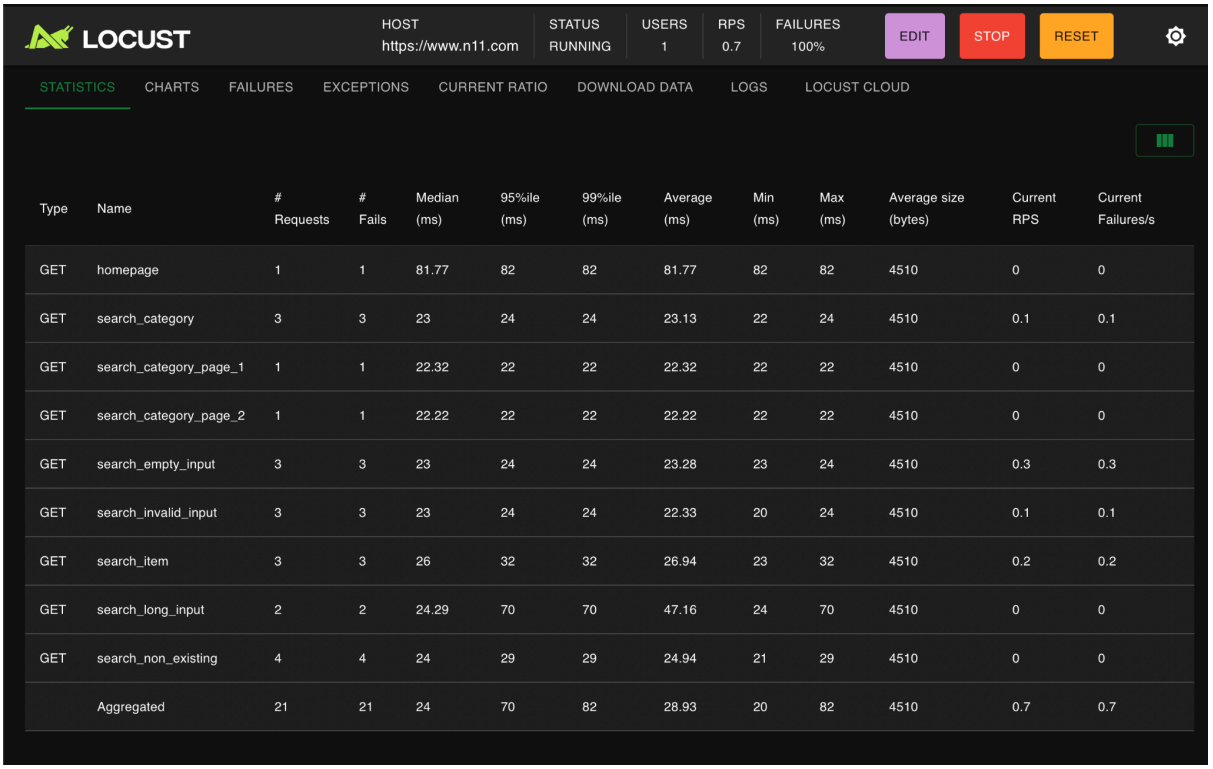
1. Go to home page.
2. Search **"araba"** and
3. Go second page: `GET /arama?q=araba&pg=2`
4. Check the response time page request

Expected Result:

1. All pages return HTTP 200
2. Average response time should be less than 2000 ms.

Test Results

Since [n11.com](n11.com) has WAF protections 403 occurs for all requests. This should work as expected in staging environment.



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	homepage	1	1	81.77	82	82	81.77	82	82	4510	0	0
GET	search_category	3	3	23	24	24	23.13	22	24	4510	0.1	0.1
GET	search_category_page_1	1	1	22.32	22	22	22.32	22	22	4510	0	0
GET	search_category_page_2	1	1	22.22	22	22	22.22	22	22	4510	0	0
GET	search_empty_input	3	3	23	24	24	23.28	23	24	4510	0.3	0.3
GET	search_invalid_input	3	3	23	24	24	22.33	20	24	4510	0.1	0.1
GET	search_item	3	3	26	32	32	26.94	23	32	4510	0.2	0.2
GET	search_long_input	2	2	24.29	70	70	47.16	24	70	4510	0	0
GET	search_non_existing	4	4	24	29	29	24.94	21	29	4510	0	0
	Aggregated	21	21	24	70	82	28.93	20	82	4510	0.7	0.7